LAB 01

INTRODUCTION TO C

Code 1:

Source Code:

```
#include <stdio.h>

main()

{

   printf("We are CSE students");

}
```

Code 2:

Source Code:

```
#include <stdio.h>
main()

{
   printf("NDUB\nCSE\nEsha\n04");
}
```

Code 3:

Source Code:

```
#include <stdio.h>

main()

{
   printf("#####\n#\n####\n#\n#");
}
```

Code 4:

Source Code:

```
#include <stdio.h>
```

```
main()
{
   printf("     #     \n");
   printf("    # #     \n");
   printf("   #   #     \n");
   printf("  #######     \n");
   printf(" #       #     \n");
   printf("#         #     \n");
}
```

Code 5:

Source Code:

```
 #include <stdio.h>

main()
{
   printf("      ########   \n");
   printf("     ###      #    \n");
   printf("    ###            \n");
   printf("   ##             \n");
   printf("  ##             \n");
   printf("   ##            \n");
   printf("    ###            \n");
   printf("     ###      #     \n");
   printf("      ########  \n");
}
```

Code 6:

Source Code:

```
#include <stdio.h>
main()
{
   printf("5*1=1 \n");
   printf("5*2=10\n");
   printf("5*3=15\n");
   printf("5*4=20\n");
   printf("5*5=25\n");
   printf("5*6=30\n");
   printf("5*7=35\n");
   printf("5*8=40\n");
   printf("5*9=45\n");
   printf("5*10=50\n");
```

```
}
```

Code 7:

Source Code:

```
#include <stdio.h>
main()
{
  printf("---------------\n");
   printf("|              |\n");
   printf("|              |\n");
   printf("|              |\n");
   printf("---------------\n");
   printf("               \n");
   printf("   ----------\n");
   printf("  /          /\n");
   printf(" /          / \n");
   printf(" ----------  \n");
 }
```

LAB 02

GENERAL PROBLEMS

Code 1:

Source Code:

```
#include <stdio.h>
int main()
{
    int num1, num2, sum;

    printf("Enter two integers: ");
    scanf("%d %d", &num1, &num2);

    sum = num1 + num2;

    printf("Summation of %d and %d is : %d", num1, num2, sum);
    return 0;
}
```

Code 2:

Source Code:

```
#include <stdio.h>
int main()
{
    float length, width, area;
    printf("Enter length of rectangle: ");
    scanf("%f", &length);
    printf("Enter width of rectangle: ");
    scanf("%f", &width);

    area = length * width;

    printf("Area of rectangle = %f sq. units", area);

   return 0;
}
```

Code 3:

Source Code:

```
#include <stdio.h>
int main()
{
    float length, area;
    printf("Enter length of square: ");
    scanf("%f", &length);


    area = length * length;

    printf("Area of square = %f sq. units", area);

    return 0;
}
```

Code 4:

Source Code:

```
#include <stdio.h>
int main()
{
    float length, width, area;
    printf("Enter length of rectangle: ");
    scanf("%f", &length);
    printf("Enter width of rectangle: ");
    scanf("%f", &width);

    area = length * width;

    printf("Area of rectangle = %f sq. units", area);

    float length2, area2;
    printf("Enter length of square: ");
    scanf("%f", &length2);


    area2 = length2 * length2;

    printf("Area of square = %f sq. units", area2);

    int num1, num2, sum;
```

```
    printf("Enter two integers: ");
    scanf("%d %d", &num1, &num2);

    sum = num1 + num2;

    printf("Summation of %d and %d is : %d", num1, num2, sum);


    return 0;
}
```

Code 5:

Source Code:

```
#include <stdio.h>
int main()
{
    int num1, num2, num3, sum;

    printf("Enter number of people in the village last year: ");
    scanf("%d", &num1);
    printf("Enter the number of people moving in: ");
    scanf("%d", &num2);
    printf("Enter the number of people moving out: ");
    scanf("%d", &num3);
    sum = (num1 + num2) - num3;

    printf("Number of people this year is : %d", num1, num2, num3, sum);
    return 0;
}
```

Code 6:

Source Code:

```
#include <stdio.h>
int main()
{
    float paid, beef, mutton, change;

    printf("Amount being paid: ");
    scanf("%f", &paid);

    printf("Kilos of beef getting bought: ");
    scanf("%f", &beef);
```

```
    printf("Kilos of mutton getting bought: ");
    scanf("%f", &mutton);
    change = paid - (beef * 700 + mutton * 800);

    printf("Amount of change is : %f", change);
    return 0;
}
```

Code 7:

Source Code:

```
#include <stdio.h>
int main()
{
    float salary, rent, expenses, savings;

    printf("Mr. Xs monthly salary is: ");
    scanf("%f", &salary);
    printf("house rent: ");
    scanf("%f", &rent);
    printf("household expenses: ");
    scanf("%f", &expenses);
    savings = salary - (rent + expenses);

    printf("Remaining money in the bank = %f", savings);
    return 0;
}
```

Code 8:

Source Code:

```
#include <stdio.h>
int main()
{
    float X, M, R;

    printf("X is: ");
    scanf("%f", &X);

    M = (X + 500)/2;
    R = X - M;

    printf("M and R are = %f, %f", M, R);
    return 0;
```

```
}
```

Code 9:

Source Code:

```
#include <stdio.h>
int main()
{
    float X, M, R;

    printf("X is: ");
    scanf("%f", &X);

    M = (X + 480)/2;
    R = X - M;

    printf("M and R are = %f, %f", M, R);
    return 0;
}
```

Code 10:

Source Code:

```
#include <stdio.h>

 int main()

{

   int x,y;

   printf("Enter the sum of age of Father & his daughter:  ");

   scanf("%d",&x);

   printf("How many times the age of the Father is the age of his daughter:  ");

   scanf("%d",&y);


   x=x/y;

   y=x*y;


   printf("Father's age is: %d\n", y);

   printf("Daughter's age is: %d\n", x);
```

```
   return 0;
}
```

Code 11:

Source Code:

```
#include <stdio.h>
 int main()
{
   float cg, c, g;
   printf("Enter the price of 2 cows and 3 goats together");
   scanf("%f",&cg);
   printf("Enter the price of 1 goat:  ");
   scanf("%f",&g);

   g=3*g;
   c=cg-g;
   c=c/2;

   printf("The price of one cow is %f: \n", c);

   return 0;
}
```

LAB 03

ARITHMETIC OPERATORS

Code 1:

Source Code:

```
#include <stdio.h>
int main()
{
        int sec, h, m, s;
        printf("Input seconds: ");
        scanf("%d", &sec);

        h = (sec/3600);

        m = (sec -(3600*h))/60;

        s = (sec -(3600*h)-(m*60));

        printf("H:M:S - %d:%d:%d\n",h,m,s);

        return 0;
}
```

Code 2:

Source Code:

```
#include <stdio.h>
int main()
{
   int days, years, weeks;
```

```
    printf("Input days: ");
            scanf("%d", &days);

    // Converts days to years, weeks and days
    years = days/365;
    weeks = (days % 365)/7;
    days = days- ((years*365) + (weeks*7));

    printf("Years: %d\n", years);
    printf("Weeks: %d\n", weeks);
    printf("Days: %d \n", days);

    return 0;
}
```

Code 3:

Source Code:

```
#include <stdio.h>
int main()
{
    int days, years, months;

    printf("Input days: ");
            scanf("%d", &days);

    // Converts days to years, months and days
    years = days/365;
    months = (days % 365)/30;
```

```
    days = days- ((years*365) + (months*30));

    printf("%d Years\n", years);
    printf("%d Months\n", months);
    printf("%d Days\n", days);

    return 0;
}
```

Code 4:

Source Code:

```
#include<stdio.h>
int main()
{
    //Program to print the average consumption of bike

    int dist, fuel;
    float avg;
    printf("Input the total distance in km: ");
    scanf("%d", &dist);
    printf("Input total fuel spent in litres: ");
    scanf("%d", &fuel);
    avg = dist / fuel;
    printf("Average consumption (km/lt) = %.2f", avg);
    return 0;
}
```

Code 5:

Source Code:

```
#include <stdio.h>
int main()
{
      int rice, consumption, days;

       printf("Input grams of rice: ");
          scanf("%d", &rice);
          printf("Input grams of rice consumed per day: ");
          scanf("%d", &consumption);

          days = rice/consumption;

          printf("Rice will be finished in: %d days", days);



   return 0;
}
```

Code 6:

Source Code:

```
#include <stdio.h>
int main()
{
         int N1, N2 , Total;

          printf("Input total number of papers: ");
             scanf("%d", &N1);
             printf("Input number of papers used to make one book: ");
             scanf("%d", &N2);
```

```
        Total = N1/N2;

        printf("Amount of books that can be made are: %d", Total);



    return 0;
}
```

Code 7:

Source Code:

```
#include <stdio.h>
int main()
{
    int employees;

   employees = 95200/800;
    printf("Number of employees in the company if the profit is 95200 Taka and each employee was given 800 Taka is: %d", employees);

      return 0;
}
```

Code 8:

Source Code:

```
 #include <stdio.h>
int main()
{
      int profit, given , employees;
```

```
        printf("Input amount of profit: ");
            scanf("%d", &profit);
            printf("Input amount given to each employee: ");
            scanf("%d", &given);

            employees = profit/given;
        printf("Number of employees in the company: %d", employees);

    return 0;
}
```

Code 9:

Source Code:

```
#include <stdio.h>
int main()
{
   int boxes;
   boxes = 43548/250;
    printf("Number of boxes necessary to pack 43548 products if every 250 products are packed in one cardboard box is: %d", boxes);

  return 0;
}
```

Code 10:

Source Code:

```
#include <stdio.h>
int main()
{
   int P1, P2, boxes;
```

```
        printf("Input amount of products: ");
          scanf("%d", &P1);
          printf("Input amount of products in each box: ");
          scanf("%d", &P2);

    boxes = P1/P2;

          printf("Number of boxes necessary to pack everything: %d", boxes);

    return 0;
}
```

Code 11:

Source Code:

```
#include <stdio.h>
int main()
{
   int E1, E2;

          E1 = 72/8;
          E2 = 15*E1;
   printf("Amount needed to pay for 15 eggs if the price of 8 eggs is 72 taka: %d taka", E2);

    return 0;
}
```

Code 12:

Source Code:

```
#include <stdio.h>
int main()
{
   int E1, E2 , E3, E4, pay;

        printf("Input amount of eggs: ");
           scanf("%d", &E1);
           printf("Input price of the given amount of eggs: ");
           scanf("%d", &E2);

           E3 = E2/E1;

           printf("Input amount of eggs being bought: ");
           scanf("%d", &E4);

           pay = E3*E4;

         printf("Amount needed to pay: %d", pay);

      return 0;
}
```

Code 13:

Source Code:

```
#include <stdio.h>
int main()
{
   int E1, E2;
         E1 = 80/4;
```

```
        E2 = 10*E1;

    printf("Amount needed to pay for 10 banans if the price of 4 bananas is 80 taka: %d taka", E2);

     return 0;
}
```

Code 14:

Source Code:

```
#include <stdio.h>
int main()
{
   int E1, E2 , E3, E4, pay;

        printf("Input amount of bananas: ");
          scanf("%d", &E1);
          printf("Input price of the given amount of bananas: ");
          scanf("%d", &E2);

          E3 = E2/E1;

          printf("Input amount of bananas being bought: ");
          scanf("%d", &E4);

        pay = E3*E4;
        printf("Amount needed to pay: %d", pay);

     return 0;
}
```

Code 15:

Source Code:

```
#include <stdio.h>
int main()
{
   int E1, E2;

            E1 = 64/8;
            E2 = 24*E1;

            printf("Amount Ayesha needs to pay for 24 pencils if 8 pencils cost 64 taka is: %d taka",
E2);



   return 0;
}
```

Code 16:

Source Code:

```
#include <stdio.h>
int main()
{
   int E1, E2 , E3, E4, pay;

         printf("Input amount of pencils: ");
            scanf("%d", &E1);
            printf("Input price of the given amount of pencils: ");
            scanf("%d", &E2);
```

```
        E3 = E2/E1;

        printf("Input amount of pencils being bought: ");
        scanf("%d", &E4);

        pay = E3*E4;
      printf("Amount needed to pay: %d", pay);

    return 0;
}
```

Code 17:

Source Code:

```
#include <stdio.h>
int main()
{
   int n, sum=0, firstdigit, lastdigit;
   printf("Enter number to find sum of first and last digit = ");
   scanf("%d", &n);

 // Find last digit of a number
   lastDigit = n % 10;

 //Find the first digit by dividing n by 10 until n greater then 10
   while(n >= 10)
   {
      n = n / 10;
   }
   firstDigit = n;
```

```
  //Calculate sum of first and last digit

    sum = firstdigit + lastdigit;
    printf("Sum of first and last digit = %d", sum);

    return 0;
}
```

LAB 04

CONDITIONAL AND OTHER OPERATORS

Code 1:

Source Code:

```
#include <stdio.h>
int main()
{
   int num1, num2, max;

   printf("Enter two numbers: ");
   scanf("%d%d", &num1, &num2);

   max = (num1 > num2) ? num1 : num2;

   printf("Maximum between %d and %d is %d", num1, num2, max);

   return 0;
}
```

Code 2:

Source Code:

```
void main()
{
        int number;

      printf("Enter a number:\t");
        scanf("%d", &number);
```

```
        (number > 0) ? printf("%d is a positive number\n", number):
        (number < 0) ? printf("%d is a negative number\n", number):
        printf("Number is zero\n");
}
```

Code 3:

Source Code:

```
void main()
{
        int number;

        printf("Enter a number:\t");
        scanf("%d", &number);

        (number%2==0) ? printf("%d is an even number\n", number): printf("%d is an odd number\n", number);

}
```

Code 4:

Source Code:

```
#include <stdio.h>
int main()
{
   int num1, num2, min;

   printf("Enter two numbers: ");
   scanf("%d%d", &num1, &num2);

   min = (num1 < num2) ? num1 : num2;
```

```
    printf("Minimum between %d and %d is %d", num1, num2, min);

    return 0;
}
```

Code 5:

Source Code:

```
#include <stdio.h>
int main()
{
   float a,b,result;
   printf("Enter two number: ");
   scanf("%f %f", &a, &b);
   result = (a>b)? a/b : a*b;
   printf("Result : %2f", result);

 return 0;
}
```

Code 6:

Source Code:

```
#include <stdio.h>
int main()
{
   int num;
   printf("Enter the number of items: ");
   scanf("%d", &num);
   (num==1) ? printf("You have 1 item") : printf("You have %d items", num);
```

```
}
```

Code 7:

Source Code:

```
#include <stdio.h>
int main()
{
    int num1, num2, num3, max;

    printf("Enter three numbers: ");
    scanf("%d%d%d", &num1, &num2, &num3);



    max = (num1 > num2 && num1 > num3) ? num1 :
          (num2 > num3) ? num2 : num3;

      printf("\nMaximum between %d, %d and %d is = %d", num1, num2, num3, max);

    return 0;
}
```

Code 8:

Source Code:

```
#include<stdio.h>
int main()
{
 int perimeter, area;
 perimeter = 2 * (5+7);
```

```
 area = 5*7;

 printf("In the given rectangle, perimeter is %d meter and area is %d square meter",
perimeter,area);
 return 0;
 }
```

Code 9:

Source Code:

```
#include<stdio.h>
int main()
{
 int num, perimeter, area;
 printf("Enter the radius of the circle: ");
 scanf("%d", &num);

 perimeter = 2 * 3.1416 * num;
 area = 3.1416 * (num*num);

 printf("In the given circle, perimeter is %d meter and area is %d square meter", perimeter,area);
 return 0;
}
```

Code 10:

Source Code:

```
#include<stdio.h>
 int main()
{
  int n;
  printf("Enter a 4 digit number\n");
```

```
  scanf("%d", &n);

 float i;
  i = (((n + 8)/3)%5)*5;
  printf("%f\n", i);
}
```

Code 11:

Source Code:

```
#include<stdio.h>
  int main()
{
  int n;

 float i, i2, i3, i4;
  printf("Enter a 4 digit number\n");
  scanf("%d", &n);

 i = n+=8;
  i2 = n/=3;
  i3 = n%=5;
  i4 = n*=5;

  printf("%f\n", i4);
}
```

Code 12:

Source Code:

```
#include <stdio.h>
void main()
{
    int int1, int2;
    printf("Input the values for Number1 and Number2 : ");
    scanf("%d %d", &int1, &int2);

 if (int1 == int2)
        printf("Number1 and Number2 are equal\n");
    else
        printf("Number1 and Number2 are not equal\n");
}
```

Code 13:

Source Code:

```
#include <stdio.h>
#include <stdlib.h>
int main()
{
    int a,b;
    printf("Enter a and b:");
    scanf("%d%d",&a,&b);

 if (a<50 && a<b)
        printf("%d<50 and %d<%d are true",a,a,b);
 else
        printf("%d<50 and %d<%d are false",a,a,b);

    return 0;
```

```
}
```

Code 14:

Source Code:

```
#include <stdio.h>
#include <stdlib.h>
int main()
{
    int a,b;
    printf("Enter a and b:");
    scanf("%d%d",&a,&b);

 if (a<50 || a<b)
        printf("At least one of %d<50 and %d<%d is true",a,a,b);

 else
        printf("%d<50 and %d<%d are false",a,a,b);

    return 0;
}
```

Code 15:

Source Code:

```
#include <stdio.h>
int main()
{
    float celsius, fahrenheit;
```

```
    printf("Enter temperature in Fahrenheit: ");
    scanf("%f", &fahrenheit);

    celsius = (fahrenheit - 32) * 5 / 9;

    printf("%.2f Fahrenheit = %.2f Celsius", fahrenheit, celsius);

    return 0;
}
```

Code 16:

Source Code:

```
#include <stdio.h>
int main()
{
      int n,l,f,m,sum,t1;

      printf("Enter 3-Digit Number: ");
      scanf("%d",&n);

      f=n/100;

      t1=n%100;

      l=t1%10;

      m=t1/10;

      printf("\nFirst Digit  = %d \nMiddle Digit = %d \nLast Digit   = %d\n",f,m,l);
```

```
    sum=l+m+f;
      printf("\nSum of All 3-Digits : %d",sum);

return 0;
}
```

LAB 05

IF, ELSE IF LADDER

Code 1:

Source Code:

```
#include <stdio.h>
int main()
 {
   int sub1, sub2, sub3;

   printf("\nEnter the marks in Mathematics: ");
   scanf("%d", &sub1);
   printf("\nEnter the marks in Physics: ");
   scanf("%d", &sub2);
   printf("Enter the marks in Chemistry: ");
   scanf("%d", &sub3);

   if((sub1>=60 && sub2>=50 && sub3>=40 && sub1+sub2+sub3>=200) || sub1+sub2>=150)
   printf("The candidate is eligible");

   else
   printf("The candidate is not eligible");

   return 0;
}
```

Code 2:

Source Code:

```
#include<stdio.h>
```

```
void main()
{
    float units,bill;
    printf("Enter the number of units consumed: ");
    scanf("%f",&units);

    if(units>=0 && units<=200)
    {
        bill = units * 0.50;
        printf("amount to be paid is: %f", bill);
    }
    else if(units>=201 && units<=400)
    {
        bill = 100 + ((units-200) * 0.65);
        printf("amount to be paid is: %f", bill);
    }
    else if(units>=401 && units<=600)
    {
        bill = 230 + ((units-400) * 0.80);
        printf("amount to be paid is: %f", bill);
    }
    else if(units>=601)
    {
        bill = 390 + ((units-600) * 1.00);
        printf("amount to be paid is: %f", bill);
        return 0;
    }
}
```

Code 3:

Source Code:

```
#include<stdio.h>
 int main()
{
  char ch;

  int amount,price;

  float net,discount,amountToBePaid;

    printf("Enter 1.If it is a pendrive purchase\nEnter 2.If it is a ram purchase\n");
    scanf("%d",&ch);

    printf("\n Enter purchased amount: ");
    scanf("%d",&amount);

    printf("Enter the price per item: ");
    scanf("%d", &price);

    net = amount * price;
    switch(ch)

 {
  case 1:
       if(amount<=100)
       {
       amountToBePaid = net;
       printf("After applying the discount, you have to pay: %0.2f", amountToBePaid);
```

```
    }
    else if(amount>=101 && amount<=200)
    {
        discount = (net*5)/100;
        amountToBePaid = net-discount;
        printf("After applying the discount, you have to pay: %0.2f", amountToBePaid);
    }
    else if(amount>=201 && amount<=300)
    {
        discount = (net*7.5)/100;
        amountToBePaid = net-discount;
        printf("After applying the discount, you have to pay: %0.2f", amountToBePaid);
    }
    else if(amount>300)
    {
        discount = (net*10)/100;
        amountToBePaid = net-discount;
        printf("After applying the discount, you have to pay: %0.2f", amountToBePaid);
    }
   break;



case 2:
    if(amount<=100)
    {
        discount = (net*5)/100;
        amountToBePaid = net-discount;
        printf("After applying the discount, you have to pay: %0.2f", amountToBePaid);
    }
```

```
        else if(amount>=101 && amount<=200)
        {
            discount = (net*7.5)/100;
            amountToBePaid = net-discount;
            printf("After applying the discount, you have to pay: %0.2f", amountToBePaid);
        }
        else if(amount>=201 && amount<=300)
        {
            discount = (net*10)/100;
            amountToBePaid = net-discount;
            printf("After applying the discount, you have to pay: %0.2f", amountToBePaid);
        }
        else if(amount>300)
        {
            discount = (net*15)/100;
            amountToBePaid = net-discount;
            printf("After applying the discount, you have to pay: %0.2f", amountToBePaid);
        }
        break;
    }
    getch();
}
```

Code 4:

Source Code:

```
#include<stdio.h>
int main()
{
    int length, breadth;
```

```
    printf("Enter the values of length and breadth:\n");
    scanf("%d%d", &length,&breadth);

    if (length==breadth)
    printf("It is a square");
    else
    printf("It is a rectangle");
    return 0;
}
```

Code 5:

Source Code:

```
#include<stdio.h>
int main()
{
    int N1, N2;

    printf("Enter the two numbers: ");
    scanf("%d%d", &N1,&N2);

    if (N1>N2)
    printf("%d is the larger number among the two", N1);
    else
    printf("%d is the larger number among the two", N2);
}
```

Code 6:

Source Code:

```
#include <stdio.h>
int main()
{
    int amount;
    int note500, note100, note50, note20, note10, note5, note2, note1;



    note500 = note100 = note50 = note20 = note10 = note5 = note2 = note1 = 0;

    printf("Enter amount: ");
    scanf("%d", &amount);

    if(amount >= 500)
    {
        note500 = amount/500;
        amount -= note500 * 500;
    }
    if(amount >= 100)
    {
        note100 = amount/100;
        amount -= note100 * 100;
    }
    if(amount >= 50)
    {
        note50 = amount/50;
        amount -= note50 * 50;
    }
    if(amount >= 20)
    {
```

```
    note20 = amount/20;
    amount -= note20 * 20;
}
if(amount >= 10)
{
    note10 = amount/10;
    amount -= note10 * 10;
}
if(amount >= 5)
{
    note5 = amount/5;
    amount -= note5 * 5;
}
if(amount >= 2)
{
    note2 = amount /2;
    amount -= note2 * 2;
}
if(amount >= 1)
{
    note1 = amount;
}

printf("Total number of notes = \n");
printf("500 = %d\n", note500);
printf("100 = %d\n", note100);
printf("50 = %d\n", note50);
printf("20 = %d\n", note20);
printf("10 = %d\n", note10);
```

```
    printf("5 = %d\n", note5);
    printf("2 = %d\n", note2);
    printf("1 = %d\n", note1);

    return 0;
}
```

LAB 06

SWITCH STATEMENT

Code 1:

Source Code:

```
#include<stdio.h>
 void main()
{
  int amount,price,ch;

  float net,discount,amountToBePaid;

     printf("enter 1.if it is a pendrive purchase\n 2.if it is a ram purchase\n");
     scanf("%d",&ch);

     printf("\n Enter purchased amount: ");
     scanf("%d",&amount);

     printf("Enter the price per item: ");
     scanf("%d", &price);

     net = amount * price;

     switch(ch)

 {
  case 1:
        if(amount<=100)
        {
```

```
 amountToBePaid = net;
 printf("After applying the discount, you have to pay: %0.2f", amountToBePaid);
 }

else if(amount>=101 && amount<=200)
{
    discount = (net*5)/100;
    amountToBePaid = net-discount;
    printf("After applying the discount, you have to pay: %0.2f", amountToBePaid);
}

else if(amount>=201 && amount<=300)
{
    discount = (net*7.5)/100;
    amountToBePaid = net-discount;
    printf("After applying the discount, you have to pay: %0.2f", amountToBePaid);
}

else if(amount>300)
{
    discount = (net*10)/100;
    amountToBePaid = net-discount;
    printf("After applying the discount, you have to pay: %0.2f", amountToBePaid);
}

break;

case 2:
    if(amount<=100)
```

```
    {
        discount = (net*5)/100;
        amountToBePaid = net-discount;
        printf("After applying the discount, you have to pay: %0.2f", amountToBePaid);
    }

    else if(amount>=101 && amount<=200)
    {
        discount = (net*7.5)/100;
        amountToBePaid = net-discount;
        printf("After applying the discount, you have to pay: %0.2f", amountToBePaid);
    }

    else if(amount>=201 && amount<=300)
    {
        discount = (net*10)/100;
        amountToBePaid = net-discount;
        printf("After applying the discount, you have to pay: %0.2f", amountToBePaid);
    }

    else if(amount>300)
    {
        discount = (net*15)/100;
        amountToBePaid = net-discount;
        printf("After applying the discount, you have to pay: %0.2f", amountToBePaid);
    }

    break;
}
```

```
    getch();

}
```

Code 2:

Source Code:

```
#include <stdio.h>
int main()
{
    int number;

    printf("Enter a positive integer number: ");
    scanf("%d",&number);

    switch(number%2) //this will return either 0 or 1
    {
        case 0:
            printf("%d is an EVEN number.\n",number);
            break;
        case 1:
            printf("%d is an ODD number.\n",number);
            break;
    }
    return 0;
}
```

Code 3:

Source Code:

```
#include <stdio.h>
int main()
```

```
{
    char ch;
    printf("Enter any alphabet: ");
    scanf("%c", &ch);

    switch(ch)
    {
        case 'a':
            printf("Vowel");
            break;
        case 'e':
            printf("Vowel");
            break;
        case 'i':
            printf("Vowel");
            break;
        case 'o':
            printf("Vowel");
            break;
        case 'u':
            printf("Vowel");
            break;
        case 'A':
            printf("Vowel");
            break;
        case 'E':
            printf("Vowel");
            break;
        case 'I':
```

```
            printf("Vowel");
            break;
        case 'O':
            printf("Vowel");
            break;
        case 'U':
            printf("Vowel");
            break;
        default:
            printf("Consonant");
    }
    return 0;
}
```

Code 4:

Source Code:

```
#include <stdio.h>
void main()
{
    int fig_code;

    float base, length, breadth, height, area, radius;

    printf("-------------------------\n");
    printf(" 1 -->Area of Circle\n");
    printf(" 2 -->Area of Rectangle\n");
    printf(" 3 -->Area of Triangle\n");
    printf("-------------------------\n");
    printf("Enter the Figure code\n");
    scanf("%d", &fig_code);
```

```
    switch(fig_code)
    {
    case 1:
        printf("Enter the radius\n");
        scanf("%f", &radius);
        area = 3.142 * radius * radius;
        printf("Area of a circle = %f\n", area);
        break;
    case 2:
        printf("Enter the breadth and length\n");
        scanf("%f %f", &breadth, &length);
        area = breadth * length;
        printf("Area of a Reactangle = %f\n", area);
        break;
    case 3:
        printf("Enter the base and height\n");
        scanf("%f %f", &base, &height);
        area = 0.5 * base * height;
        printf("Area of a Triangle = %f\n", area);
        break;

    default:
        printf("Error in figure code\n");
        break;
    }
}
```

Code 5:

Source Code:

```
#include<stdio.h>
int main()
{
        int years,months,days,hours,minutes,seconds;

        printf("\n Enter the number of years: ");
        scanf("%d",&years);
       months = 12 * years;

      days = 365 * years;
      hours = 24 * 365 * years;
      minutes = 60 * 24 * 365 * years;
      seconds = 60 * 60 * 24 * 365 * years;
       printf("\n %d Months",months);
       printf("\n %d Days",days);
       printf("\n %d Hours",hours);
       printf("\n %d minutes",minutes);
       printf("\n %d seconds",seconds);
     return 0;
}
```

LAB 07

NESTED LOOPS

Code 1:

Source Code:

```
#include<stdio.h>
 int main()
{
  int i;

  for(i=1;i<=10;i++)
  printf("%d\t", i);

}
```

Code 2:

Source Code:

```
#include<stdio.h>
int main()
{
  int i,even=0;

  printf("\nEven numbers between 1 to 10 are:");

  for(i=1;i<=10;i++)
  {
    if(i%2==0)
    {
      printf("\n%d",i);
```

```
      even++;
    }
  }
  return 0;
}
```

Code 3:

Source Code:

```
#include <stdio.h>
int main()
{
    int i;

     printf("All odd numbers from 1 to 10 are: \n", i);

    for(i=1; i<=10; i++)
    {
        if(i%2!=0)
        {
            printf("%d\n", i);
        }
    }
    return 0;
 }
```

Code 4:

Source Code:

```
#include<stdio.h>
int main()
```

```
{
  int i,even=0;

  for(i=1;i<=10;i++)
  {
    if(i%2==0)
    {
       even++;
    }
  }

  printf("\nTotal even numbers between 1 to 10 :%d",even);

  return 0;
}
```

Code 5:

Source Code:

```
#include<stdio.h>
int main()
{
  int i,odd=0;

  for(i=1;i<=10;i++)
  {
    if(i%2!=0)
    {
       odd++;
    }
```

```
  }

  printf("\nTotal odd numbers between 1 to 10 :%d",odd);

  return 0;
}
```

Code 6:

Source Code:

```
#include <stdio.h>
int main()
{
    int  j, sum = 0;

    for (j = 1; j <= 5; j++)
    {
        sum = sum + j;
    }

    printf("\nThe Summation of 1 to 5  is : %d\n", sum);
}
```

Code 7:

Source Code:

```
#include <stdio.h>
int main()
{
    int i, sum=0;
```

```
    for(i=2; i<=10; i+=2)
    {
        sum += i;
    }

    printf("Sum of all even number between 1 to 10 are = %d",sum);

    return 0;
}
```

Code 8:

Source Code:

```
#include <stdio.h>
int main()
{
    int i, n, sum=0;

    for(i=1; i<=10; i+=2)
    {
        sum += i;
    }

    printf("Sum of odd numbers = %d", sum);
    return 0;
}
```

Code 9:

Source Code:

```
#include<stdio.h>
```

```
int main()
{
    int val;

    printf("1. All numbers between 1 to 10 using loop\n2. Even numbers between 1 to 10\n3. Odd numbers between 1 to 10.\n4. Count of even between 1 to 10\n5. Count of odd between 1 to 10.\n6. Summation of 1 to 5\n7. Summation of even numbers between 1 to 10\n8. Summation of odd numbers between 1 to 10\n");

    printf("\n Press any number from 1 to 8");
    scanf("%d",&val);

    switch (val)
{
 case 1:
 {
  int i;
  for(i=1;i<=10;i++)

  printf("%d\t", i);
}

 break;

 case 2:
 {
  int i,even=0;

  printf("\nEven numbers between 1 to 10 are:");
```

```
  for(i=1;i<=10;i++)
  {
    if(i%2==0)
    {
      printf("\n%d",i);
      even++;
    }
  }
  return 0;
}

 break;

 case 3:
 {
    int i;

     printf("All odd numbers from 1 to 10 are: \n", i);

    for(i=1; i<=10; i++)
    {
        if(i%2!=0)
        {
            printf("%d\n", i);
        }
    }
    return 0;
}
```

```
 break;

 case 4:
 {
  int i,even=0;

  for(i=1;i<=10;i++)
  {
    if(i%2==0)
    {
       even++;
    }
  }

  printf("\nTotal even numbers between 1 to 10 :%d",even);

  return 0;
}

 break;

 case 5:
 {
  int i,odd=0;

  for(i=1;i<=10;i++)
  {
    if(i%2!=0)
    {
```

```
        odd++;
    }
  }

  printf("\nTotal odd numbers between 1 to 10 :%d",odd);

  return 0;
}

 break;

 case 6:
 {
    int  j, sum = 0;

    for (j = 1; j <= 5; j++)
    {
        sum = sum + j;
              }
    printf("\nThe Summation of 1 to 5  is : %d\n", sum);
}

 break;

 case 7:
 {
    int i, sum=0;

    for(i=2; i<=10; i+=2)
```

```
    {
        sum += i;
    }

    printf("Sum of all even number between 1 to 10 are = %d",sum);

    return 0;
}

 break;

 case 8:
 {
    int i, n, sum=0;

    for(i=1; i<=10; i+=2)
    {
        sum += i;
    }
    printf("Sum of odd numbers = %d", sum);

    return 0;
}

 break;
 default:
 printf("input error!!");
}
}
```

LAB 08

ARRAY

Code 1:

Source Code:

```
#include <stdio.h>
int main() {

  int values[5];

  printf("Enter 5 integers: ");

  for(int i = 0; i < 5; ++i)
 {
     scanf("%d", &values[i]);
  }
  printf("Displaying integers: ");

  for(int i = 0; i < 5; ++i){

     printf("%d\n",values[i]);
  }
  return 0;
}
```

Code 2:

Source Code:

```
#include <stdio.h>
```

```
int main()
{
  int values[5];

  printf("Enter 5 integers: ");

  for(int i = 0; i < 5; ++i)
 {
     scanf("%d", &values[i]);
  }

  printf("Displaying the squares of the given values: ");

  for(int i = 0; i < 5; ++i)
{
    values[i] = values[i] * values[i];

     printf("%d\n",values[i]);
  }
  return 0;
}
```

Code 3:

Source Code:

```
#include<stdio.h>
int main()
{
    int arr[10], i;
```

```
    printf("Enter any 10 array elements: ");

    for(i=0; i<10; i++)
    scanf("%d", &arr[i]);
    printf("\nAll Even Array elements are:\n");

    for(i=0; i<10; i++)
    {
        if(arr[i]%2==0)
        {
            printf("%d ", arr[i]);
        }
    }
    return 0;
}
```

Code 4:

Source Code:

```
#include<stdio.h>
int main()
{
    int arr[10], i;

    printf("Enter any 10 array elements: ");

    for(i=0; i<10; i++)
    scanf("%d", &arr[i]);
    printf("\nAll Odd Array elements are:\n");
```

```
    for(i=0; i<10; i++)
    {
        if(arr[i]%2==1)
        {
            printf("%d ", arr[i]);
        }
    }
    return 0;
}
```

Code 5:

Source Code:

```
#include<stdio.h>
int main()
{
    int a[10],i,even=0;

    printf("Enter any 10 Array elements: ");

    for(i=0; i<10; i++)
    {
        scanf("%d", &a[i]);
    }
        for(i=0; i<10; i++)
    {
        if(a[i]%2==0)
        even++;
    }
    printf("Total number of even integers among the given values are: %d", even);
```

```
    return 0;
}
```

Code 6:

Source Code:

```
#include<stdio.h>
int main()
{
    int a[10],i,odd=0;

   printf("Enter any 10 Array elements: ");

    for(i=0; i<10; i++)
      {
        scanf("%d", &a[i]);
     }
        for(i=0; i<10; i++)
    {
        if(a[i]%2==1)
        odd++;
    }
    printf("Total number of odd integers among the given values are: %d", odd);

    return 0;
}
```

Code 7:

Source Code:

```
#include<stdio.h>
 main()
 {
   int a[5],b[5],i,total[5];

   printf("\nEnter elements of first array: ");

   for(i=0;i<5;i++)
   {
   scanf("%d",&a[i]);
   }
   printf("\nEnter elements of second array: ");

   for(i=0;i<5;i++)
   {
   scanf("%d",&b[i]);
   }
   for(i=0;i<5;i++)
   {
   total[i]=a[i]+b[i];
   }
   printf("\nSummation of the elements among the arrays: ");

   for(i=0;i<5;i++)
   {
   printf("\n  \t\t\t %d",total[i]);
   }
 }
```

Code 8:

Source Code:

```
#include<stdio.h>

int main()

{

    int val;


    printf("1. Display array elements\n2. Display the squares of the given values\n3. All even array elements among the given integers.\n4. All odd elements among the given integers.\n5. Total number of even integers among the given values.\n6. Total number of odd integers among the given values\n7. Summation of the elements among the arrays\n");


    printf("\n Press 1 to 7");

    scanf("%d",&val);


    switch (val)

{

    case 1:

     {


  int values[5];


  printf("Enter 5 integers: ");


  for(int i = 0; i < 5; ++i)

 {

     scanf("%d", &values[i]);

  }
```

```
  printf("Displaying integers: ");
  for(int i = 0; i < 5; ++i){

     printf("%d\n",values[i]);
  }
  return 0;
}

 break;

 case 2:
 {

  int values[5];

  printf("Enter 5 integers: ");

  for(int i = 0; i < 5; ++i)
{
     scanf("%d", &values[i]);
  }

  printf("Displaying the squares of the given values: ");

  for(int i = 0; i < 5; ++i)
{
    values[i] = values[i] * values[i];
     printf("%d\n",values[i]);
  }
```

```
  return 0;
}

    break;

    case 3:
    {
    int arr[10], i;

    printf("Enter any 10 array elements: ");

    for(i=0; i<10; i++)
        scanf("%d", &arr[i]);
    printf("\nAll Even Array elements are:\n");

    for(i=0; i<10; i++)
    {
        if(arr[i]%2==0)
        {
            printf("%d ", arr[i]);
        }
    }
    return 0;
}
    break;

    case 4:
    {
    int arr[10], i;
```

```
    printf("Enter any 10 array elements: ");

    for(i=0; i<10; i++)
     scanf("%d", &arr[i]);
    printf("\nAll Odd Array elements are:\n");

    for(i=0; i<10; i++)
    {
        if(arr[i]%2==1)
        {
            printf("%d ", arr[i]);
        }
    }
    return 0;
}
    break;

    case 5:
    {
    int a[10],i,even=0;
    printf("Enter any 10 Array elements: ");

    for(i=0; i<10; i++)
    {
          scanf("%d", &a[i]);
    }
          for(i=0; i<10; i++)
    {
```

```
        if(a[i]%2==0)
        even++;
    }
      printf("Total number of even integers among the given values are: %d", even);

    return 0;
}
    break;

    case 6:
    {
    int a[10],i,odd=0;

    printf("Enter any 10 Array elements: ");

    for(i=0; i<10; i++)
    {
     scanf("%d", &a[i]);
   }
     for(i=0; i<10; i++)
   {
        if(a[i]%2==1)
        odd++;
    }
    printf("Total number of odd integers among the given values are: %d", odd);

    return 0;
}
    break;
```

```
 case 7:
 {
int a[5],b[5],i,total[5];

printf("\nEnter elements of first array: ");

for(i=0;i<5;i++)
{
scanf("%d",&a[i]);
}
printf("\nEnter elements of second array: ");

for(i=0;i<5;i++)
{
scanf("%d",&b[i]);
}
for(i=0;i<5;i++)
{
total[i]=a[i]+b[i];
}
printf("\nSummation of the elements among the arrays: ");

for(i=0;i<5;i++)
{
printf("\n  \t\t\t %d",total[i]);

}
 }
```

```
    break;

}
}
```

----------O----------